

PÉTER LODRI

SOTA AI SYSTEMS ARCHITECT · TECHNICAL AI SAFETY · QUANTITATIVE SYSTEMS RESEARCHER

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TARGET FOCUS & PROFILE SUMMARY

Independent AI researcher and systems architect specializing in **empirical multi-agent verification**, **mathematical AI safety bounds**, and **low-bit neural consensus distillation**. Proven track record of architecting zero-reward-hacking evaluation harnesses, deriving exact geometric-mean consensus equivalences, and deploying low-latency edge models with formal spectral guarantees. Deeply aligned with Coefficient Giving's mission to address catastrophic risks from transformative AI through rigorous epistemic foundations, open-source verification tools, and pragmatic systems engineering.

SELECTED RESEARCH & OPEN SYSTEMS CONTRIBUTIONS

KOMPRESS v2: Geometric-Mean Consensus Distillation & Spectral Rigidity

2025 – 2026

Lead Author & Architect · kompress.vaked.dev/paper/main_v2.pdf · HF Model

- **Consensus Equivalence Theorem:** Proved that arithmetic logit averaging across teacher councils induces exact normalized geometric-mean probability consensus $\bar{p} \propto \prod p_k^{1/K}$, and that minimizing mean reverse KL to teachers is optimization-equivalent to minimizing reverse KL against $\bar{p}$.
- **Spectral Rigidity & Null Models:** Formulated Marchenko-Pastur spectral edges ($\lambda_{\pm} = p(1 \pm \sqrt{\gamma})$) for ternary $\{-1, 0, +1\}$ BitLinear weight matrices, establishing restricted lower/upper singular bounds ($0 < c \leq |W_x| \leq C$) on active semantic subspaces to prevent dimensional collapse.
- **Empirical Results:** Improved edge student validation cross-entropy from 2.1369 to 1.6120, reduced memory from 3.4 GB to 0.42 GB (\$8.1\times\$), and accelerated Apple Silicon inference to 118 tokens/s (\$4.2\times\$).

DeepSiper Enthea: Worktree-Isolated Multi-Agent Safety & AST Verifier

2025 – 2026

Principal Architect · github.com/8b-is/deepsiper-enthea

- **Zero Reward-Hacking Verification:** Designed a multi-agent evaluation harness coordinating parallel reasoners across isolated Git worktrees, replacing textual proxy metrics with an explicit 4-predicate admissibility contract: $\mathcal{A}(c) = \mathcal{P}(c) \wedge \mathcal{S}(c) \wedge \mathcal{T}(c) \wedge \mathcal{G}(c)$ (Parse, AST Invariant, Test Suite, Git Sandbox).
- **Empirical Benchmark Sweep:** Achieved 100.0% pass rate across 5 algorithmic test suites (AST logic, $O(\log N)$ matrix power, 256-bit XOR Kademlia distance, BitLinear quantization), demonstrating zero adversarial escapes under the verifier suite.
- **Modular Cordis Microkernel:** Engineered 51 independent packages supporting automated subagent dispatch, live AST inspections, and deterministic replay traces.

Axiom Quant: Pure Mathematics, Random Matrices & Stochastic Process Substrate

2024 – 2026

Founding Researcher · axiomquant.org · 86 Monographs

- **Mathematical AI Safety:** Authored extensive monographs on Erdős threshold phase transitions, Schramm-Loewner Evolution (SLE), Tracy-Widom edge distributions, and stochastic repair bounds in autonomous systems.
- **Parametricity & Commitment Honesty:** Applied Reynolds' relational parametricity and Wadler's free theorems to prove that polymorphic agent skill delivery ($\text{RESOLVE} \text{ } \text{head}$) can operate at zero standing prompt token overhead, eliminating architectural commitment leakage.

EtherHive & Honest-IRC: Sovereign Post-Quantum Cryptographic Messaging

2024 – 2026

Lead Cryptographic Engineer · etherhive.vaked.dev · github.com/peterlodri-sec/etherhive

- **Post-Quantum Primitives:** Implemented FIPS 203/204/205 standards (ML-KEM-1024 / Kyber-1024, ML-DSA-87 / Dilithium, SLH-DSA) integrated with X25519 Double Ratchet forward secrecy.

- **Decentralized Epistemic Identity:** Pioneered `honesty-auth` , deriving sovereign cryptographic secrets from relational invariants rather than centralized KYC or OAuth authorities.

TECHNICAL CORE COMPETENCIES

AI Safety & Eval:	Multi-agent worktree harnesses, AST invariant validation, reward-hacking prevention, mechanistic interpretability, empirical incident deconstruction, free-theorem verification.
Modeling & Distill:	Geometric-mean logit consensus, teacher council ensembles, BitNet b1.58 ternary quantization ($\{-1, 0, +1\}$), PyTorch, MLX (Apple Silicon Metal kernels), Hugging Face Hub.
Mathematics:	Random matrix theory (Marchenko-Pastur, Tracy-Widom), stochastic calculus, spectral graph theory, Hawkes processes, Reynolds parametricity, category-theoretic semantics.
Systems & Infra:	Rust (async, zero-copy, SIMD, no-GIL), Python 3.13 JIT, TypeScript, Cloudflare Workers/Durable Objects, Kademia DHT, Git worktrees, Linux eBPF/memfd.

EDUCATION & PHILOSOPHICAL FOUNDATIONS

Self-Directed Higher Studies in Applied Mathematics & Computer Science	Continuous
Deep study in random matrix theory, stochastic differential equations, abstract algebra, and category theory. Epistemic focus on Cartesian skepticism, falsifiability (Popper/Lakatos), and the causal coupling of artificial cognition to physical substrates.	